

Implement Client Script & UI Policy (Incident)

Phase 6: Project Documentation

Item	Details
Project Name	Implement Client Script & UI Policy (Incident)
Platform	ServiceNow
Target Table	Incident [incident]
Team ID	SWTID-2026-7176
Document Type	Solution overview, configuration guide, user guide and test record

1. Project Overview

1.1 Purpose

Incident records need consistent and accurate data for triage, routing, SLA tracking and reporting. Relying on user awareness alone leads to incomplete or inconsistent records. This project enforces the rules at the user interface level with a UI Policy and Client Scripts, so records meet the business rules before they are submitted.

1.2 Scope

- **In scope:** the Incident form and the Incident list view in the desktop UI of ServiceNow.
- **Out of scope:** API, import and other server-side updates, and tables other than Incident.

1.3 Skills Demonstrated

Incident Management, UI Policy, UI Policy Actions, Client Scripts and Form Validation.

2. Solution Summary

2.1 Business Rules

ID	Business Rule
BR1	When Impact is 1 - High, Assignment group must be provided.
BR2	When Impact is 1 - High, Urgency is set to High and cannot be edited.
BR3	When Impact is 1 - High, an incident cannot be saved unless Assigned to is filled in.
BR4	When Impact is no longer High, the restrictions applied by the UI Policy are reversed.
BR5	State cannot be changed by editing in the list view; it is changed on the Incident form.

2.2 Components and Traceability

Component	Type	What It Does	Rule
High Impact Control	UI Policy	Applies when Impact is 1 - High and reverses when it is not	BR1, BR4
Assignment group action	UI Policy Action	Sets Assignment group to Mandatory	BR1
Urgency action	UI Policy Action	Sets Urgency to Read only	BR2

Component	Type	What It Does	Rule
Auto set urgency for high impact	onChange Client Script	Sets Urgency to 1 (High) and shows a message when Impact changes to High	BR2
Prevent save if Assigned To missing	onSubmit Client Script	Blocks the save and shows an error on Assigned to	BR3
Prevent state change via list edit	onCellEdit Client Script	Shows an alert and cancels the State edit in the list	BR5

2.3 Incident Fields Used

Label	Field Name	Type	Used By
Impact	impact	Choice	UI Policy condition, onChange, onSubmit
Urgency	urgency	Choice	UI Policy Action, onChange
Assignment group	assignment_group	Reference	UI Policy Action
Assigned to	assigned_to	Reference	onSubmit
State	state	Choice	onCellEdit

3. Configuration Guide

Log in to the instance with administrative or configuration access before starting.

3.1 Create the UI Policy

1. Go to **System UI** → **UI Policies** and click **New**.
2. Set **Table** to Incident, the **Short description** to High Impact Control, and **Active** to true.
3. Under **When to Apply**, set the condition **Impact is 1 - High**.
4. Check **Reverse if false** so the policy undoes its changes when the condition is not met.
5. Click **Submit**.

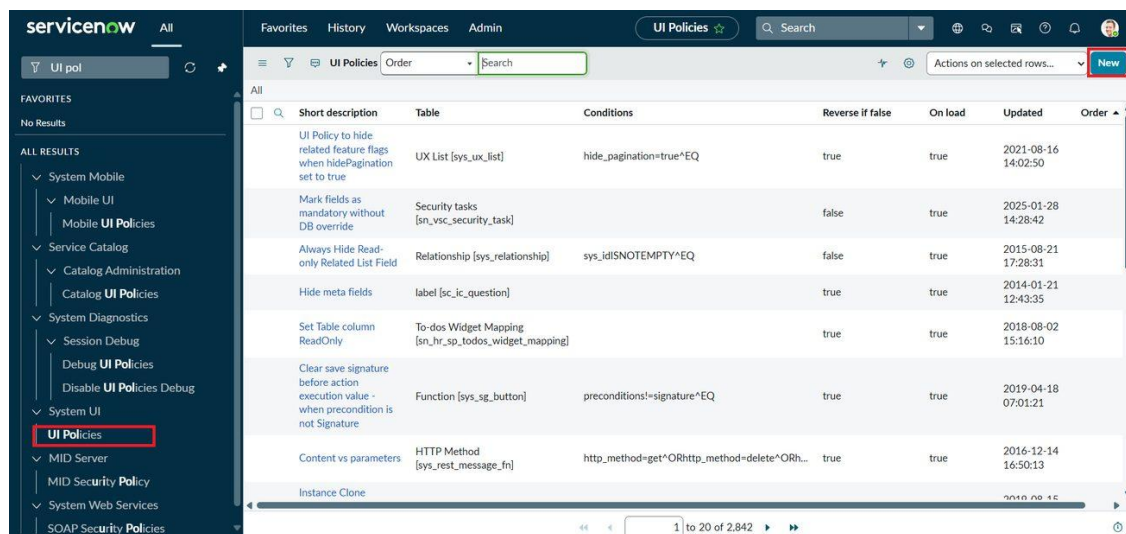


Figure 1: UI Policies list with the New button

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UI Policy - High Impact Control

UI Policy High Impact Control View: Advanced

Table Incident [Incident]

Application Global

Active ☒

* Short description High Impact Control

Order 100

When to Apply Script

Conditions Add Filter Condition Add OR Clause

Impact is 1 - High

AND OR

If selected, the UI Policy applies to all form views; otherwise the UI Policy is view-specific

Global ☒

Reverse the effects of the UI policy actions when the Conditions evaluate to false

Reverse if false ☒

Apply the UI policy actions when the form is loaded and when the user changes values on the form

On load ☒

Tables that extend the specified Table inherit this UI Policy

Inherit ☐

Figure 2: High Impact Control UI Policy on the Incident table

3.2 Add the UI Policy Actions

1. In the High Impact Control policy, open the **UI Policy Actions** related list and click **New**.
2. For the first action, set **Field name** to Assignment group and **Mandatory** to True.
3. For the second action, set **Field name** to Urgency, **Read only** to True and leave **Visible** as Leave alone.
4. Click **Submit** for each action.

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UI Policy Action - assignment_group

UI Policy Action assignment_group

UI policy High Impact Control

Table Incident [Incident]

Field name Assignment group

Mandatory True

Visible Leave alone

Read only False

Clear the field value ☐

Update Delete

Figure 3: UI Policy Action: Assignment group is Mandatory

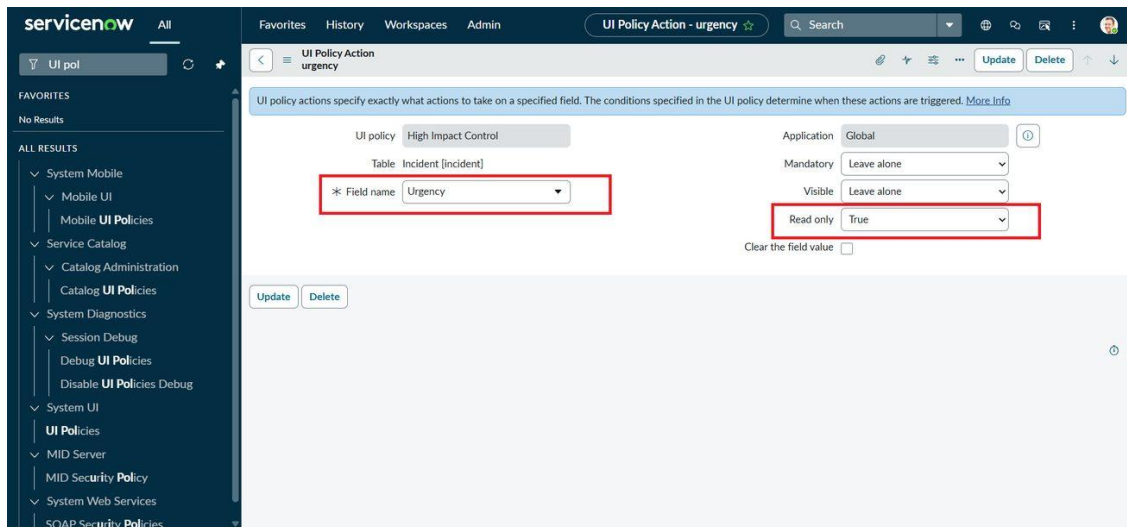


Figure 4: UI Policy Action: Urgency is Read only

3.3 Create the Client Scripts

Go to **System UI** → **Client Scripts**, click **New**, and create the three scripts below. All three use Table = Incident, UI Type = Desktop and Active = true.

Name	Type	Field name	Purpose
Auto set urgency for high impact	onChange	Impact	Set Urgency to High when Impact becomes High
Prevent save if Assigned To missing	onSubmit	Not used	Block the save when Assigned to is empty for High impact
Prevent state change via list edit	onCellEdit	State	Cancel State edits made in the list

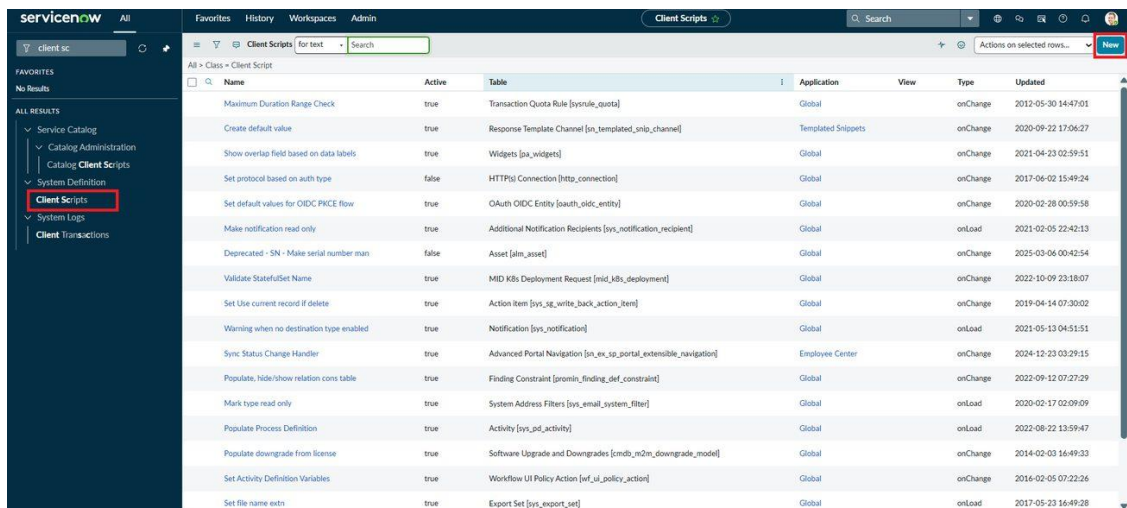


Figure 5: Client Scripts list with the New button

Client Script - Auto set urgency for high impact

Name: Auto set urgency for high impact
 Table: Incident [Incident]
 UI Type: Desktop
 Type: onChange
 Field name: Impact

Application: Global
 Active: ☒
 Inherited: ☐
 Global: ☒

Description:

Messages:

Script:

```

1 function onChange(control, oldValue, newValue, isLoading) {
2   if (isLoading || newValue === '') {
3     return;
4   }
5   if (newValue == "1") { // High Impact
6     g_form.setValue('urgency', '1');
7     g_form.showErrorMessage(
8       'Impact',
9       'High Impact incidents must have High urgency and an assignment group.',
10      'info'
11    );
12  }
13 }
14
15

```

Isolate script: ☒

Update Delete

Figure 6: onChange script: Auto set urgency for high impact

Client Script - Prevent save if assigned to missing

Name: Prevent save if assigned to missing
 Table: Incident [Incident]
 UI Type: Desktop
 Type: onSubmit

Application: Global
 Active: ☒
 Inherited: ☐
 Global: ☒

Description:

Messages:

Script:

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4     g_form.showErrorMessage(
5       'assigned_to',
6       'Assigned to is mandatory for High Impact Incidents.'
7     );
8     return false;
9   }
10  return true;
11 }
12
13
14

```

Isolate script: ☒

Update Delete

Figure 7: onSubmit script: Prevent save if Assigned To missing

Client Script - Prevent state change via list edit

Name: Prevent state change via list edit
 Table: Incident [Incident]
 UI Type: Desktop
 Type: onCellEdit
 Field name: State

Application: Global
 Active: ☒
 Inherited: ☐
 Global: ☒

Description:

Messages:

Script:

```

1 function onCellEdit(sysId, table, oldValues, newValue, callback) {
2
3   alert('State cannot be changed from the list. Please open the Incident.');
```

```

4   // Cancel the list edit
5   callback(false, oldValues);
6 }
7
8
9

```

Isolate script: ☒

Update Delete

Figure 8: onCellEdit script: Prevent state change via list edit

3.4 Script Reference

onChange (Impact)

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') {
        g_form.setValue('urgency', '1');
        g_form.addInfoMessage('Urgency set to High for High impact incident.');
```

onSubmit

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorMessage(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}
```

onCellEdit (State)

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
    alert('State cannot be changed from the list. Please open the Incident.');
```

The guide gives two versions of the onChange and onCellEdit scripts (text and screenshot). This document uses the text version for onChange and the screenshot version for onCellEdit.

4. User Guide

This section describes what an agent sees when working with Incident records.

When the User...	The System...	Rule
Sets Impact to 1 - High	Makes Assignment group mandatory, sets Urgency to High, makes Urgency read-only and shows an information message	BR1, BR2
Clicks Submit with Assigned to empty (Impact High)	Blocks the save and shows "Assigned To is mandatory for High impact incidents." under Assigned to	BR3
Fills in Assigned to and clicks Submit	Saves the incident	BR3
Changes Impact from High to another value	Removes the mandatory and read-only settings so the fields can be edited again	BR4
Double-clicks State in the list view	Shows an alert and keeps the State value unchanged	BR5
Changes State on the Incident form and clicks Update	Saves the change	BR5

5. Testing Record

The tests follow Task 6 of the project guide. The screenshots below show the observed behaviour.

No.	Test	Expected Result	Evidence
T1	Save with Impact High and Assigned to empty	Save blocked with an error on Assigned to	Figure 9
T2	Save with Assigned to filled	Incident saves without errors	Figure 10
T3	Change Impact from High to Medium	Mandatory and read-only rules are removed	Figure 11
T4	Edit State in the list view	Alert appears and State is unchanged	Figure 12
T5	Change State on the form	State change saves	Figures 13 and 14

The screenshot shows the 'Incident - Create' form in ServiceNow. The form is titled 'Incident - Create INC0010012'. The 'Assigned to' field is highlighted with a red box, and a red error message is displayed below it: 'Assigned To is mandatory for High Impact Incidents.' The 'Impact' field is set to '1 - High'. The 'Short description' field contains 'testing UI policy'.

Figure 9: T1: Save blocked with an error on Assigned to

The screenshot shows the 'Incident - Create' form in ServiceNow. The 'Assigned to' field is now filled with 'Beth Anglin' and is highlighted with a red box. The 'Impact' field is still set to '1 - High'. The 'Short description' field contains 'testing UI policy'.

Figure 10: T2: Assigned to filled in, ready to submit

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Incident - INC0010012

Number: INC0010012

Channel: -- None --

State: New

Impact: 2 - Medium

Urgency: 1 - High

Priority: 2 - High

Assignment group: Software

Assigned to:

* Short description: testing UI policy

Description:

Country: -- None --

States: -- None --

Figure 11: T3: Impact changed to 2 - Medium, Assigned to no longer required

servicenow All

dev354432.service-now.com says
State cannot be changed from the list. Please open the Incident.

Number	Opened	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2024-01-14 02:15:14	Inquiry / Help	Software	(empty)	2024-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	Software	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	Software	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	Inquiry / Help	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment: ATF Assessor	Software	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF: Test1	Software	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	Inquiry / Help	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	Software	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	Hardware	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Inquiry / Help	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Software	(empty)	2016-08-10	admin

Figure 12: T4: Alert shown when State is edited in the list

servicenow All

Incident - INC0008111

Number: INC0008111

Channel: -- None --

State: In Progress

Impact: 3 - Low

Urgency: 3 - Low

Priority: 5 - Planning

Assignment group:

Assigned to:

* Short description: ATF: Test1

Description:

Country: -- None --

States: -- None --

Save

- Add to Visual Task Board
- Copy Incident
- Create Incident Task
- Create Outage
- Create Problem
- Create Request
- Create Child Incident
- Create Normal Change
- Create Standard Change
- Create Emergency Change
- Refresh Impacted Services
- Metrics Timeline
- Follow on Live Feed
- Analyze Access
- Configure
- Export
- View
- Create Favorite
- Copy URL
- Copy sys_id
- Show XML
- History
- Reload form

Figure 13: T5: State changed to In Progress on the form

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:31:59	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:23	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Photo Placeholder	3 -	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin

Figure 14: T5: List view showing the saved State

6. Known Limitations and Troubleshooting

Situation	Likely Cause	What to Check
Urgency stays at High after Impact is lowered	The onChange script sets Urgency only when Impact becomes High and does not reset it	Change Urgency by hand; the field is editable again after the reverse
Save is not blocked for a High impact incident	The onSubmit script is inactive, or the incident is updated through an API or import	Confirm the script is Active and that the update comes from the form
Assignment group is not mandatory	The UI Policy is inactive, Reverse if false is misconfigured, or the condition is wrong	Check Active and the condition Impact is 1 - High
State can be edited in the list without an alert	The onCellEdit script is inactive or the field name is not State	Check Active, Type onCellEdit and Field name State
Scripts do not run on a different UI	The scripts use UI Type Desktop	Create matching scripts for other UIs if they are needed

7. Maintenance and Future Enhancements

- **Maintenance:** keep every record Active and named as in Section 2.2, and change configuration only with administrative access.
- **Server-side enforcement:** add a Business Rule or Data Policy so the rules also apply to API and import updates.
- **Urgency reset:** set Urgency back to a default value when Impact is lowered.
- **Consistent rules:** decide whether Assignment group, Assigned to or both should be required, and align the UI Policy and the onSubmit script.

8. Conclusion

The project shows how a UI Policy and three Client Scripts work together to keep Incident data complete and consistent. The configuration is small, uses only standard ServiceNow features and was verified with five tests. This document records the design, setup, use and testing of the solution, so it can be repeated, maintained and extended.